Alberto Salvarese

Email: alberto.salvarese@utexas.edu LinkedIn: Alberto Salvarese

Profile

Physics Ph.D. candidate and Distinguished Graduate Fellow at UT Austin developing statistical models and CPU/GPU-accelerated Python and JAX pipelines for noisy observational and simulated data. Experienced in formulating hypotheses, designing rigorous simulation-based tests, comparing alternative models, and drawing statistically sound conclusions. Four first-author papers and active member of the LIGO-Virgo-KAGRA Collaboration.

Education

- **Ph.D. in Physics**, University of Texas at Austin *Aug. 2022 – Present*
  Expected graduation: 12/2027 | GPA: 3.98/4.0 | Advisor: Prof. Hsin-Yu Chen
  Focus: gravitational-wave astrophysics, Bayesian hierarchical inference, population modeling, and data analysis
- **M.Sc. in Astrophysics and Cosmology**, Università di Padova *Oct. 2019 – Apr. 2022*
  Thesis: Probing Parity Violation in the Early Universe | Final Grade: 110/110 cum laude
- **B.Sc. in Astronomy**, Università di Padova *Oct. 2016 – Sept. 2019*
  Thesis: Constraints on Neutron Star EoS from Multiwavelength GW170817 Data | Final Grade: 107/110

Quantitative Research Experience

Doctoral Researcher - Gravitational-Wave Astrophysics, UT Austin *Jul. 2023 – Present*

- Developed end-to-end Python and JAX pipelines for Bayesian hierarchical inference on simulated and observational datasets, using CPU/GPU acceleration to study population models and astrophysical parameters
- Designed repeated-simulation studies to quantify model discrimination, parameter uncertainty, selection effects, and sensitivity to measurement precision.
- Developed and validated bias-mitigation methods that reduced systematic error below statistical uncertainty; forecasted measurement performance for the next decade of observations.
- Led projects from problem formulation through implementation, validation, publication, and conference presentation within an international scientific collaboration.

Selected Quantitative Projects

- Bayesian Time-Series Modeling of Epidemiological Data (R/JAGS): implemented SIR/SEIR models on real COVID-19 data; performed Bayesian inference, uncertainty quantification, and model comparison using DIC.
- Machine Learning for Student Exam Outcomes (Python): built preprocessing and feature-engineering pipelines; tuned regression, Gaussian-process, and ensemble models; evaluated performance by cross-validation and interpreted predictions with SHAP.
- Thermal Modeling for CMB-S4: characterized cryogenic Al-6061 behavior through physical measurements and numerical modeling; designed and machined test specimens using SolidWorks.

Research output & Recognition

Publications: 4 first-author papers (2 peer-reviewed, 2 under review) **Presentations:** Oral presentations at APS 2024, 2025, and 2026, and EDSU Tool 2024. **Programs:** Erdős Institute Quantitative Finance Program (2026, ongoing); selected participant at Caltech and the Institute for Advanced Study

Technical Skills

Programming: Python (JAX, NumPy, SciPy, Pandas, scikit-learn, Astropy, Matplotlib), R, Git, Linux, Jupyter, CPU/GPU and high-performance computing, LaTeX.
Methods: Bayesian inference, hierarchical modeling, Monte Carlo methods, nested sampling, machine learning, uncertainty quantification, simulation, cross-validation, and model comparison.